

Logistic Regression

Data

```
URL <- "https://raw.githubusercontent.com/DS4PS/pe4ps-textbook/master/data/admissions.csv"
```

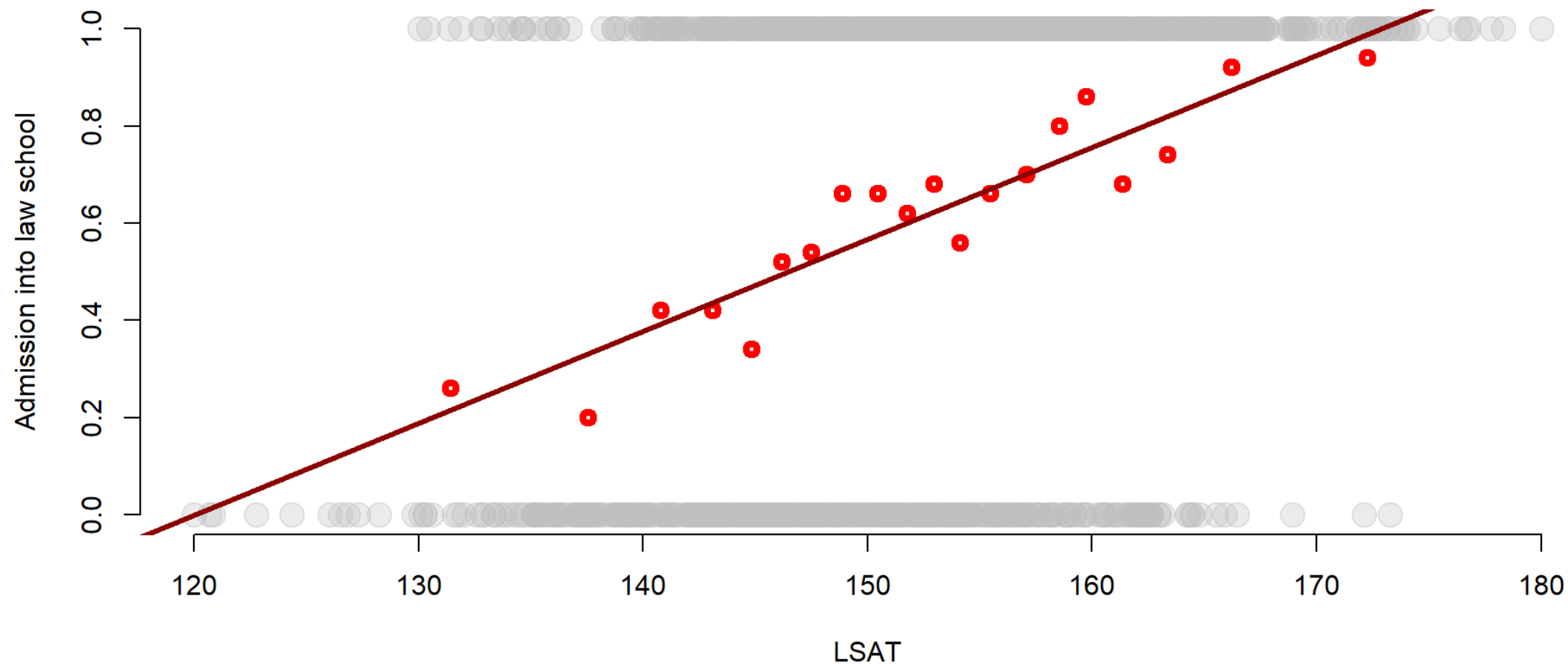
```
dat <- read.csv( URL, stringsAsFactors=F )
```

Predicting Law School Admissions:

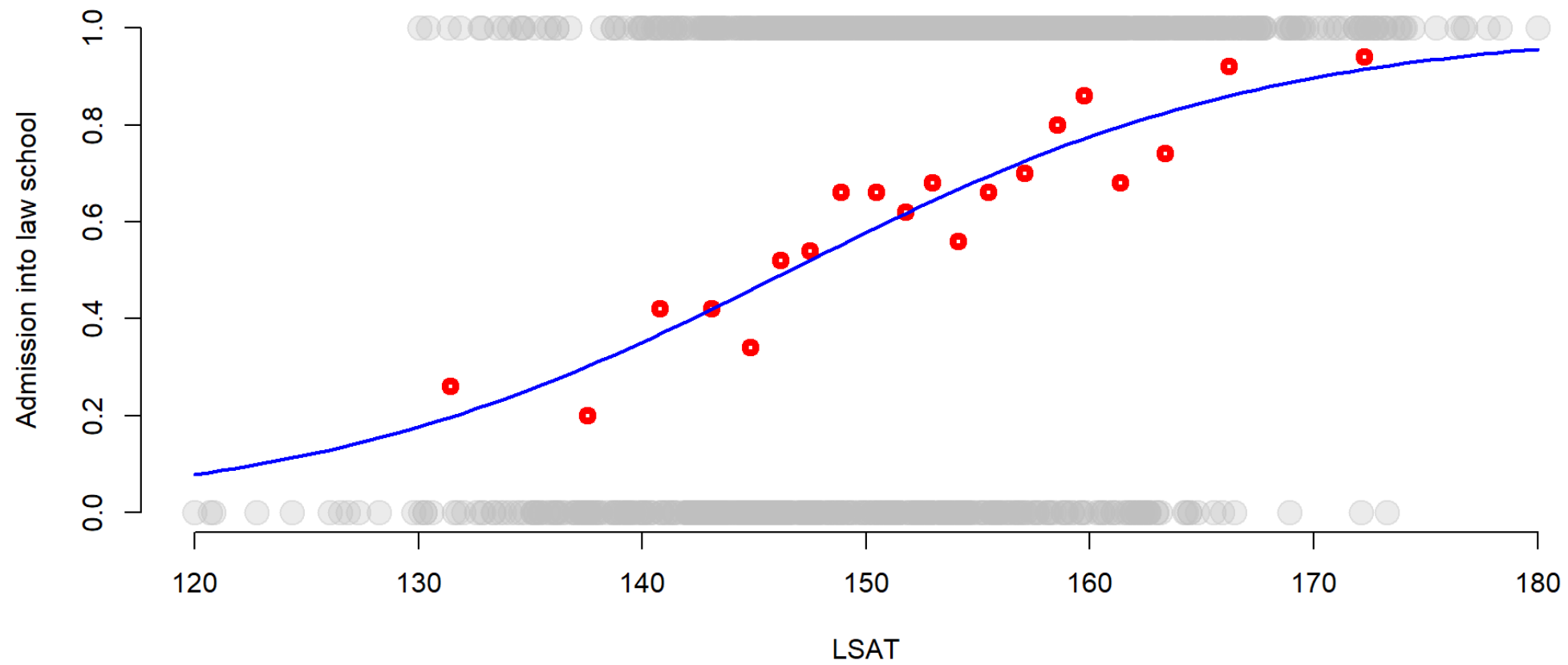
$$Y = \beta_0 + \beta_1 * LSAT + \beta_1 * Essay + \beta_1 * GPA + e$$

Variable name	Description
LSAT	LSAT score (from 120 to 180)
Essay	Essays score (from 1 to 365)
GPA	Average GPA (from 0 to 5)
Admission	The student has been admitted (=1) or not (=0)

Linear Probability Model:



Logit Model:

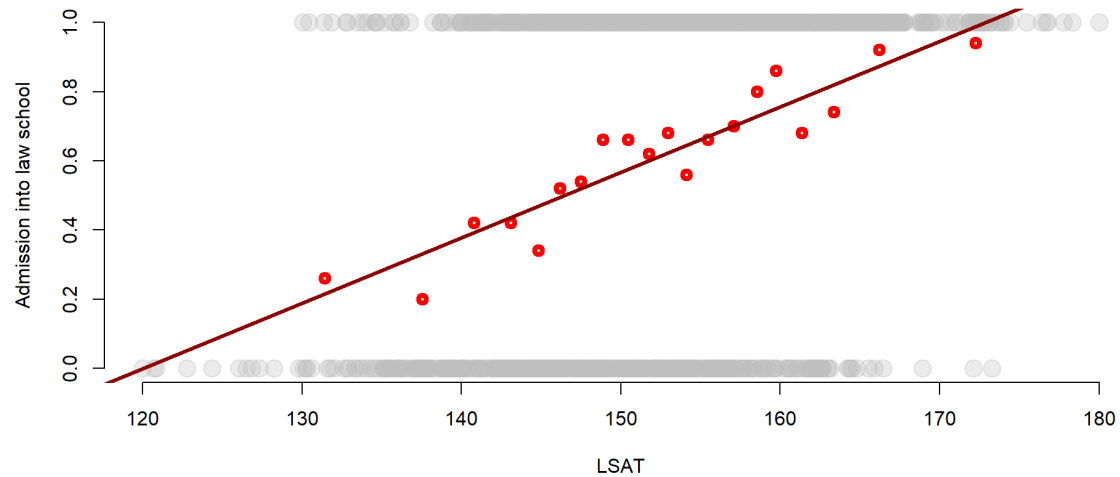


Interpretation:

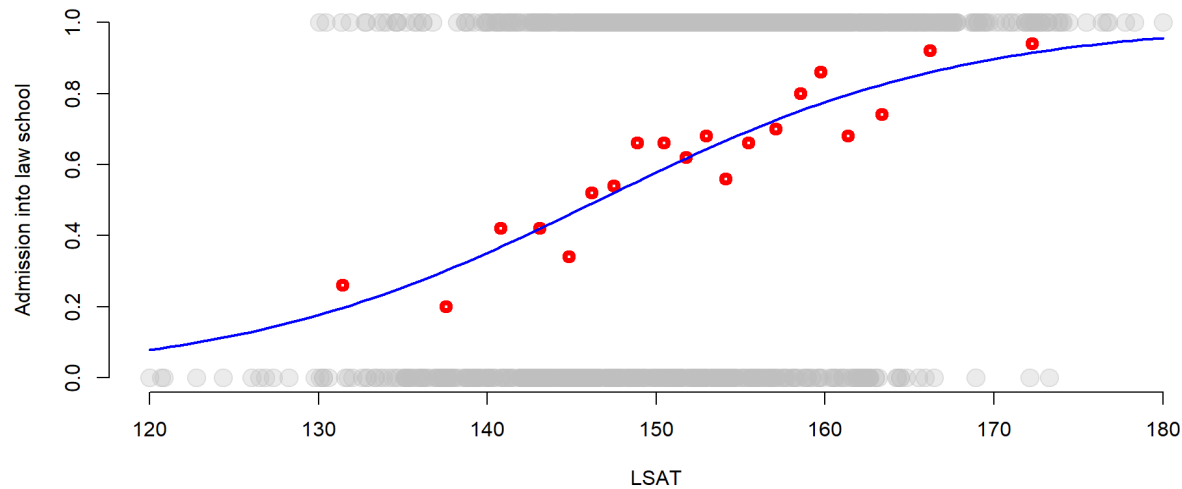
$$p = \frac{1}{1 + e^{-(\beta_0 + x \cdot \beta)}}$$

Logit link function:

Translates regression coefficients
to probabilities.



Both models
would produce
similar predictions.



The predictions
would differ most
in the tails.

```

URL <- "https://raw.githubusercontent.com/D8498/pw4ps-textbook/master/data/admissions.csv"
dat <- read.csv( URL, stringsAsFactors=F )

head( dat )

### LINEAR PROBABILITY MODEL

m1 <- lm( Admission ~ LSAT + Essay + GPA, data=dat )
summary( m1 )

# Marginal effects comparison:
#
# The standard deviation is a "reasonable"
# approximation of a big improvement.
#
# Which of these three things should I spend
# time on to improve my likelihood of admissions?

sd.lsat <- sd( dat$LSAT )
sd.essay <- sd( dat$Essay )
sd.gpa <- sd( dat$GPA )

b0 <- m1$coefficients[1]
b1 <- m1$coefficients[2]
b2 <- m1$coefficients[3]
b3 <- m1$coefficients[4]

sd.lsat * b1      # gains from improvement in LSAT
sd.essay * b2     # gains from improvement in essay
sd.gpa * sd.gpa   # gains from improvement in gpa

# predicted prob of success (admissions) for a specific individual:

gpa <- 3.0
lsat <- 145
essay <- 90

b0 + b1*lsat + b2*essay + b3*gpa

```

```

### LOGIT MODEL

m2 <- glm( Admission ~ LSAT + Essay + GPA, data = dat, family = "binomial" )
summary( m2 )

```

```

# LOGIT LINK FUNCTION

p = 1 / ( 1 + exp( - ( b0 + b1*lsat + b2*ss2 ) ) )

```

```

# predicted prob of success (admissions) for a specific individual:

```

```

b0 <- m2$coefficients[1]
b1 <- m2$coefficients[2]
b2 <- m2$coefficients[3]
b3 <- m2$coefficients[4]

gpa <- 3.0
lsat <- 145
essay <- 90

# Model reports log-odds
y.hat <- b0 + b1*lsat + b2*essay + b3*gpa

# convert log-odds to probabilities
1 / ( 1 + exp( - ( y.hat ) ) )

# Marginal effects comparison:
#
# We must "center" other variables
# to look at marginal effects of one (LSAT here):

gpa <- mean( dat$GPA )
essay <- mean( dat$Essay )
lsat <- 150

y.hat <- b0 + b1*lsat + b2*essay + b3*gpa

1 / ( 1 + exp( - ( y.hat ) ) )

```

```

# Expected gains change greatly depending
# on the baseline, eg LSAT of 130 vs 150 vs 170

# value of 10-point increase in LSAT
# when current score is a 140

```

```

gpa <- mean( dat$GPA )
essay <- mean( dat$Essay )

lsat <- 140
y.hat <- b0 + b1*lsat + b2*essay + b3*gpa
p.140 <- 1 / ( 1 + exp( - ( y.hat ) ) )

lsat <- 150
y.hat <- b0 + b1*lsat + b2*essay + b3*gpa
p.150 <- 1 / ( 1 + exp( - ( y.hat ) ) )

p.150 - p.140

```

```

# value of 10-point increase in LSAT
# when current score is a 120

lsat <- 120
y.hat <- b0 + b1*lsat + b2*essay + b3*gpa
p.120 <- 1 / ( 1 + exp( - ( y.hat ) ) )

lsat <- 130
y.hat <- b0 + b1*lsat + b2*essay + b3*gpa
p.130 <- 1 / ( 1 + exp( - ( y.hat ) ) )

p.130 - p.120

```

```

# value of 10-point increase in LSAT
# when current score is a 170

```

```

lsat <- 170
y.hat <- b0 + b1*lsat + b2*essay + b3*gpa
p.170 <- 1 / ( 1 + exp( - ( y.hat ) ) )

```